

## 2.1 · Electric dipole radiation

### Lorentz model · Classical radiation in vacuum

[Lorentz model](#) · [The Lorentz oscillator](#)

Picture two opposite charges moving relative to each other, or as a simple special case, one charge moving while the opposite charge remains fixed at the origin. Their electric field cannot rearrange everywhere at once, the changing dipole moment produces a time-dependent electromagnetic field, part of which travels outward as radiation. On this page, we will follow that wave from the moving dipole to its fields, energy flux, and total radiated power. **SI units** are used throughout.

#### Geometry and assumptions

Consider a localized, electrically neutral source with dipole moment

$$\mathbf{p}(t) = p(t)\hat{\mathbf{z}}. \quad (1)$$

In Equation 1, the dipole points along the  $z$  axis. Use spherical coordinates  $(r, \theta, \phi)$  centered on the source:  $\theta$  is measured from the positive  $z$  axis, and  $\phi$  is the azimuthal angle. The unit vectors  $\hat{\mathbf{r}}$ ,  $\hat{\theta}$ , and  $\hat{\phi}$  point in the directions of increasing  $r$ ,  $\theta$ , and  $\phi$ . The two identities we will need are

$$\hat{\mathbf{z}} = \cos \theta \hat{\mathbf{r}} - \sin \theta \hat{\theta}, \quad \hat{\mathbf{r}} \times \hat{\theta} = \hat{\phi}. \quad (2)$$

[Download the editable SVG](#). It can be edited with Inkscape.

The geometry in Figure 1 makes Equation 2 visible: the three local unit vectors rotate as the field point moves. For two charges  $+q$  and  $-q$ , let  $\mathbf{d}(t)$  point from the negative charge to the positive charge. Then  $\mathbf{p}(t) = q\mathbf{d}(t)$ .

We use the **electric-dipole approximation**: the maximum charge separation  $d_{\max}$  is much smaller than the radiation wavelength, with  $kd_{\max} \ll 1$ , and the observation distance satisfies  $r \gg d_{\max}$ . The wave number in this condition is defined by

### Dipole and spherical coordinate unit vectors

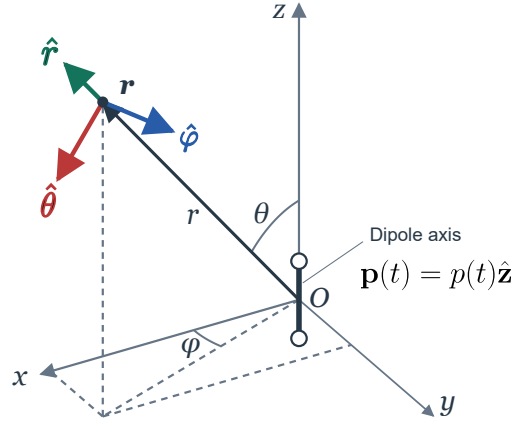


Figure 1: Spherical coordinates of an electric dipole. At the field point  $\mathbf{r}$ , the unit vectors  $\hat{\mathbf{r}}$ ,  $\hat{\theta}$ , and  $\hat{\phi}$  point in the directions of increasing  $r$ ,  $\theta$ , and  $\phi$ . They form a right-handed basis:  $\hat{\theta} \times \hat{\phi} = \hat{\mathbf{r}}$ . Arrow lengths are schematic.

$$k = \frac{\omega}{c} = \frac{2\pi}{\lambda}. \quad (3)$$

Equation Equation 3 tells us how rapidly the phase changes in space. The fields at time  $t$  cannot know the source's present motion; they depend on its earlier motion at the **retarded time**

$$t_r = t - \frac{r}{c}. \quad (4)$$

The delay  $r/c$  in Equation 4 is simply the travel time from source to observer. Dots denote time derivatives. Unless stated otherwise,  $p$ ,  $\dot{p}$ , and  $\ddot{p}$  in the field equations are evaluated at  $t_r$ .

### Brief derivation from retarded potentials

You can skip the derivation

### Dipole potentials

The retarded potentials of a charge density  $\rho$  and current density  $\mathbf{J}$  are

$$\Phi(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0} \int \frac{\rho(\mathbf{r}', t - R/c)}{R} d^3r', \quad \mathbf{A}(\mathbf{r}, t) = \frac{\mu_0}{4\pi} \int \frac{\mathbf{J}(\mathbf{r}', t - R/c)}{R} d^3r', \quad (5)$$

In Equation 5,  $R = |\mathbf{r} - \mathbf{r}'|$  is the distance from a source point to the observer. Expanding these potentials for a small neutral source and using  $\int \mathbf{J} d^3r' = \dot{\mathbf{p}}$  gives

$$\Phi = \frac{\cos \theta}{4\pi\epsilon_0} \left[ \frac{p(t_r)}{r^2} + \frac{\dot{p}(t_r)}{cr} \right], \quad \mathbf{A} = \frac{\mu_0}{4\pi r} \dot{p}(t_r) \hat{\mathbf{z}}. \quad (6)$$

Thus  $A_r = \mu_0 \dot{p} \cos \theta / (4\pi r)$ ,  $A_\theta = -\mu_0 \dot{p} \sin \theta / (4\pi r)$ , and  $A_\phi = 0$ .

### Take the derivatives

Now comes the central step: turn the potentials in Equation 6 into measurable fields using

$$\mathbf{E} = -\nabla\Phi - \frac{\partial\mathbf{A}}{\partial t}, \quad \mathbf{B} = \nabla \times \mathbf{A}. \quad (7)$$

Be careful when differentiating. Changing  $r$  also changes the retarded time in Equation 4, so

$$\frac{\partial p(t_r)}{\partial r} = -\frac{\dot{p}(t_r)}{c}. \quad (8)$$

Using Equation 7 and Equation 8, the transverse electric field becomes

$$E_\theta = -\frac{1}{r} \frac{\partial\Phi}{\partial\theta} - \frac{\partial A_\theta}{\partial t} = \frac{\sin \theta}{4\pi\epsilon_0} \left[ \frac{p}{r^3} + \frac{\dot{p}}{cr^2} + \frac{\ddot{p}}{c^2 r} \right]. \quad (9)$$

This one calculation, Equation 9, already reveals the three distance dependences  $1/r^3$ ,  $1/r^2$ , and  $1/r$ . Applying the same derivatives to  $E_r$  and  $B_\phi$  gives the remaining components.

## The dipole-field solution

The complete electric-dipole fields outside the dipole source are:

$$\mathbf{E}(r, \theta, t) = \frac{1}{4\pi\epsilon_0} \left\{ 2 \cos \theta \left[ \frac{p}{r^3} + \frac{\dot{p}}{cr^2} \right] \hat{\mathbf{r}} + \sin \theta \left[ \frac{p}{r^3} + \frac{\dot{p}}{cr^2} + \frac{\ddot{p}}{c^2 r} \right] \hat{\theta} \right\}. \quad (10)$$

$$\mathbf{B}(r, \theta, t) = \frac{\mu_0 \sin \theta}{4\pi} \left[ \frac{\dot{p}}{r^2} + \frac{\ddot{p}}{cr} \right] \hat{\phi}. \quad (11)$$

Equation 10 and Equation 11 contain every nonzero component:  $E_\phi = B_r = B_\theta = 0$ . There is no dependence on  $\phi$  because rotating the system around the dipole axis changes nothing.

| Distance dependence | Interpretation                                         |
|---------------------|--------------------------------------------------------|
| $r^{-3}$            | Quasistatic electric-dipole field                      |
| $r^{-2}$            | Induction terms                                        |
| $r^{-1}$            | Radiation fields that carry a finite power to infinity |

Far from the source,  $kr \gg 1$ , the  $1/r$  terms dominate over the faster-decaying near-field terms. Keeping only those terms in Equation 10 and Equation 11 gives the so called Radiation zone solution.

## Radiation zone and harmonic solution

For observation point far from the source,  $kr \gg 1$ , the solution of field is

$$\mathbf{E}_{\text{rad}}(\mathbf{r}, t) = \frac{1}{4\pi\epsilon_0 c^2} \frac{\hat{\mathbf{r}} \times [\hat{\mathbf{r}} \times \ddot{\mathbf{p}}(t_r)]}{r},$$

$$\mathbf{B}_{\text{rad}} = \frac{1}{c} \hat{\mathbf{r}} \times \mathbf{E}_{\text{rad}}. \quad (12)$$

For a dipole directed along  $z$ ,  $\ddot{\mathbf{p}}(t) = \ddot{p}(t) \hat{\mathbf{z}}$ . Because

$$\hat{\mathbf{r}} \times (\hat{\mathbf{r}} \times \hat{\mathbf{z}}) = \sin \theta \hat{\theta}$$

We have

$$\boxed{\mathbf{E}_{\text{rad}} = \frac{\ddot{p}(t_r) \sin \theta}{4\pi\epsilon_0 c^2 r} \hat{\theta}}, \quad \boxed{\mathbf{B}_{\text{rad}} = \frac{\ddot{p}(t_r) \sin \theta}{4\pi\epsilon_0 c^3 r} \hat{\phi}}. \quad (13)$$

Equation 13 shows that both fields are perpendicular to  $\hat{\mathbf{r}}$  and therefore transverse to the propagation direction.

To see an actual traveling wave, let us study a harmonic dipole, which oscillates sinusoidally with **peak amplitude**  $p_0$ :

$$p(t) = p_0 \cos \omega t, \quad \ddot{p}(t_r) = -\omega^2 p_0 \cos(\omega t - kr). \quad (14)$$

Substituting Equation 14 into Equation 13 gives

$$\mathbf{E}_{\text{rad}}(r, t) = -\frac{\omega^2 p_0 \sin \theta}{4\pi\epsilon_0 c^2 r} \cos(\omega t - kr) \hat{\theta}, \quad (15)$$

$$\mathbf{B}_{\text{rad}}(r, t) = \frac{E_{\theta}(r, t)}{c} \hat{\phi}. \quad (16)$$

Equations Equation 15 and Equation 16 share the phase  $\omega t - kr$ , the signature of an outward-traveling wave.

### Far-field Poynting vector

The Poynting vector is the electromagnetic energy flux, measured in  $\text{W m}^{-2}$ :

$$\mathbf{S} = \frac{1}{\mu_0} \mathbf{E} \times \mathbf{B}. \quad (17)$$

and for a linearly polarized sinusoidal wave, its one cycle average is

$$\langle |\mathbf{S}| \rangle = \frac{E_0^2}{2c\mu_0} = \frac{cE_0^2}{2\epsilon_0}$$

Insert the radiation fields from Equation 13 into Equation 17. Because  $\hat{\theta} \times \hat{\phi} = \hat{\mathbf{r}}$ , the energy flows outward:

$$\boxed{\mathbf{S}_{\text{rad}}(r, \theta, t) = \frac{[\ddot{p}(t_r)]^2 \sin^2 \theta}{16\pi^2 \epsilon_0 c^3 r^2} \hat{\mathbf{r}}.} \quad (18)$$

For the harmonic dipole, the rapid oscillations are usually not resolved. Averaging Equation 18 over one period and using  $\langle \cos^2(\omega t - kr) \rangle = 1/2$  gives

$$\boxed{\langle \mathbf{S}_{\text{rad}}(r, \theta, t) \rangle = \frac{p_0^2 \omega^4 \sin^2 \theta}{32\pi^2 \epsilon_0 c^3 r^2} \hat{\mathbf{r}}.} \quad (19)$$

Read the  $\sin^2 \theta$  factor in Equation 19 physically: radiation is strongest sideways, in the equatorial plane, and vanishes along the dipole axis. The pattern is shaped like a doughnut. The intensity falls as  $1/r^2$ , exactly compensating for the growing area of a sphere.

Figure 2 shows an SVG drawing of the angular distribution, corresponding to Eq. (14).

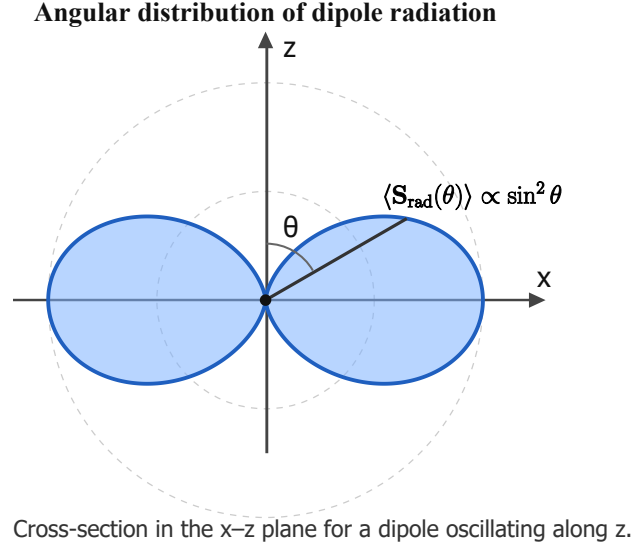


Figure 2: Dipole-radiation angular distribution. At each polar angle  $\theta$ , the distance from the origin to the blue curve is proportional to the radiated power per unit solid angle,  $d\langle P \rangle / d\Omega$ .

### Angular distribution and total radiated power

An element of solid angle is  $d\Omega = \sin \theta d\theta d\phi$ , and it cuts out an area  $r^2 d\Omega$  on a sphere of radius  $r$ . Multiplying the radial flux in Equation 18 by this area gives

$$dP = |\mathbf{S}_{\text{rad}}| dA = \frac{[\ddot{p}(t_r)]^2 \sin^2 \theta}{16\pi^2 \epsilon_0 c^3} d\Omega. \quad (20)$$

To collect the radiation sent in every direction, integrate the angular factor in Equation 20:

$$\int \sin^2 \theta d\Omega = \int_0^{2\pi} d\phi \int_0^\pi \sin^3 \theta d\theta = 2\pi \times \frac{4}{3} = \frac{8\pi}{3}. \quad (21)$$

Using Equation 21, the power crossing a large sphere at  $r$  is

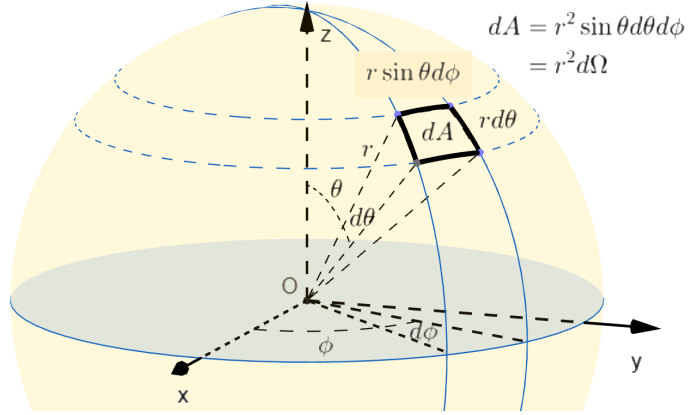


Figure 3: An element of solid angle  $d\Omega$  and its corresponding surface element  $dA$  on a sphere of radius  $r$ .

$$P_r(t) = \int dP = \frac{[\ddot{p}(t - r/c)]^2}{6\pi\epsilon_0 c^3}. \quad (22)$$

The retarded time in Equation 22 matters: the sphere receives now what the source emitted  $r/c$  earlier. Equivalently, the instantaneous source power is  $P_{\text{emit}}(t) = \ddot{p}(t)^2/(6\pi\epsilon_0 c^3)$ .

For the harmonic dipole,

$$\langle P \rangle = \frac{p_0^2 \omega^4}{12\pi\epsilon_0 c^3}. \quad (23)$$

The factor  $1/2$  from time averaging is essential because  $p_0$  is a peak amplitude, not an RMS value. Equation 23 contains a striking result: at fixed dipole amplitude, 20% of frequency increase give a power increase of  $1.2^4 \approx 2$ .

### Connection to the Lorentz electron model

For an electron displaced by  $z(t)$  relative to a fixed positive center,

$$p(t) = -ez(t), \quad \ddot{p}(t) = -e\ddot{z}(t), \quad (24)$$

where  $e > 0$  is the elementary charge. The emitted power becomes the nonrelativistic Larmor result for linear motion:

$$P_{\text{emit}}(t) = \frac{e^2 \ddot{z}(t)^2}{6\pi\epsilon_0 c^3}. \quad (25)$$

Substituting Equation 24 into Equation 22 gives Equation 25. An oscillating bound electron therefore reradiates energy, and its acceleration—not merely its displacement—sets the instantaneous power.

## Summary

Harmonic dipole radiation can be reduced to three main results.

The far-zone electric field is

$$\mathbf{E}_{\text{rad}}(r, t) = -\frac{p_0 \omega^2 \sin \theta}{4\pi\epsilon_0 c^2 r} \cos(\omega t - kr) \hat{\theta},$$

The angular distribution of the time-averaged radiated power is

$$\frac{d\langle P \rangle}{d\Omega} = \frac{p_0^2 \omega^4}{32\pi^2 \epsilon_0 c^3} \sin^2 \theta.$$

The total time-averaged radiated power is

$$\langle P \rangle = \frac{p_0^2 \omega^4}{12\pi\epsilon_0 c^3}.$$

The essential physical result is

$$E_{\text{rad}} \propto \frac{\ddot{p}}{r},$$

so that

$$P_{\text{rad}} \propto \ddot{p}^2 \propto \omega^4 p_0^2.$$

## Check your understanding

1. Why is there no radiation along the dipole axis?
2. Why can a field proportional to  $1/r$  carry the same total power through spheres of different radii?
3. If  $p_0$  is unchanged and the frequency doubles, how does the mean radiated power change?



### Further reading

- MIT, *Electromagnetic Fields and Energy*, §12.2: Electrodynamic fields of a current loop and electric dipole.
- The Feynman Lectures on Physics, Vol. II, Chapter 21: Solutions of Maxwell's equations with currents and charges.